

▪ Writing Do ... While | Until Statement

The Do ... While | Until Statement is used to specify that a set of statements should repeat as long as the condition specified is True. This type of Do ... While | Until Statement which checks for a condition after the Statements have executed once.

The syntax for the Do ... While | Until statement is as follows:

Do

Statement(s)

Exit Do

Loop While | Until Condition

Example 1:

```
Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
    Dim Counter As Integer = 0
    Do
        MessageBox.Show("Value is :" & Counter)
        Counter = Counter + 2
    Loop While Counter <= 10
End Sub
```

Example2:

Design the following form and write a program in VB. Net to read ten integer numbers then find the Odd and Even number between them using Do ... Until Statement and ComboBox?

The Solution is:

To design the above form we implement the following set properties of the objects:

Object	Property	Setting
Button1	Text	Find
Button2	Text	Clear
Button3	Text	Exit
Label1	Text	The Odd Numbers
Label2	Text	The Even Numbers
Form1	Text	Odd and Even Numbers
Form1	MaximizeBox	False
Form1	MinimizeBox	False
ComboBox1		
ComboBox2		

Writing the Code

Use the Code Editor

1. Double-click the Find Button1 on the form and write the following code inside it:

```
Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
    Dim X As Integer
    Dim I As Integer = 0
    Do
        X = InputBox("Enter The 10 Numbers")
        If X Mod 2 = 0 Then
            ComboBox2.Items.Add(X)
        Else
            ComboBox1.Items.Add(X)
        End If
        I = I + 1
    Loop Until I >= 10
End Sub
```

2. Double-click the Clear Button2 on the form and write the following code inside it:

```
Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
    ComboBox1.Items.Clear()
    ComboBox2.Items.Clear()
End Sub
```

3. Double-click the Exit Button3 on the form and write the following code inside it:

```
Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click
    End
End Sub
```

▪ Writing Do ... While Statement

The Do ... While Statement is used to specify that a set of statements should repeat as long as the condition specified is True. This type of Do ... While Statement which checks for a condition before the executing the Loop.

The syntax for the Do ... While statement is as follows:

Do While | Until Condition

Statement(s)

Exit Do

Loop

Example 1:

```
Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
    Dim Counter As Integer = 1
    Do Until Counter >= 10
        MessageBox.Show("Value is :" & Counter)
        Counter = Counter + 2
    Loop
End Sub
```

Example2:

Design the following form and write a program in VB. Net to read ten integer numbers then find the Odd and Even number between them using Do ... While Statement and ComboBox?

The Solution is:

To design the above form we implement the following set properties of the objects:

Object	Property	Setting
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Form1	Text	Odd and Even Numbers
Form1	MaximizeBox	False
Form1	MinimizeBox	False
ComboBox1		
ComboBox2		

Writing the Code

Use the Code Editor

1. Double-click the Find Button1 on the form and write the following code inside it:

```
Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
    Dim X As Integer
    Dim I As Integer = 0
    Do While I < 10
        X = InputBox("Enter The 10 Numbers")
        If X Mod 2 = 0 Then
            ComboBox2.Items.Add(X)
        Else
            ComboBox1.Items.Add(X)
        End If
        I = I + 1
    Loop
End Sub
```

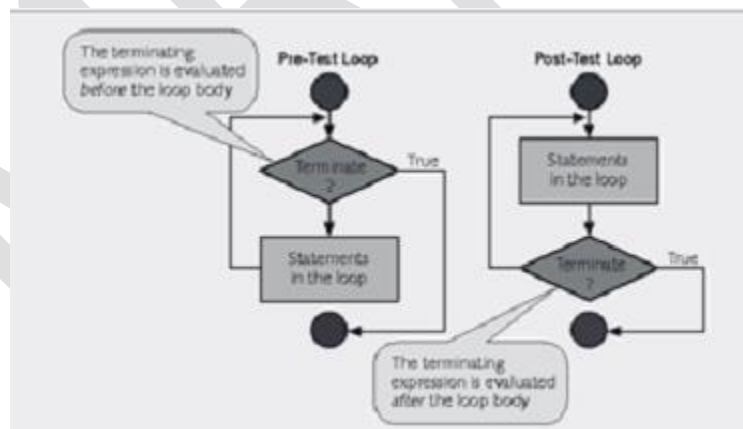
2. Double-click the Clear Button2 on the form and write the following code inside it:

```
Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
    ComboBox1.Items.Clear()
    ComboBox2.Items.Clear()
End Sub
```

3. Double-click the Exit Button3 on the form and write the following code inside it:

```
Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click  
    End  
End Sub
```

- Difference between a Do While and Do Until loop:
 - Do While loop executes *while* the expression is true.
 - Do Until loop executes *until* the expression is false.
- **Writing Post-Test Loops**
- Programming languages provide two kinds of loops:
 - Pre-test loop tests the terminating condition at the *beginning* of the loop.
 - Post-test loop tests the terminating condition at the *end* of the loop.



Loop Structures

- Do While and Do Until loops can be written as either pre-test or post-test loops.
- For Next and While loops are always pre-test.

▪ Nested loop:

- A loop within a loop.
- Can be constructed using any combination of Do While, Do Until, or For Next loops.

Example:

```
Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
    Dim X As Integer
    Dim I As Integer = 0
    Dim J As Integer
    Do While I < 3
        X = InputBox("Enter The 10 Numbers")
        For J = 1 To 3
            X = X - 1
            If X Mod 2 = 0 Then
                ComboBox2.Items.Add(X)
            Else
                ComboBox1.Items.Add(X)
            End If
        Next
        I = I + 1
    Loop
End Sub
```